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The Communication Challenge

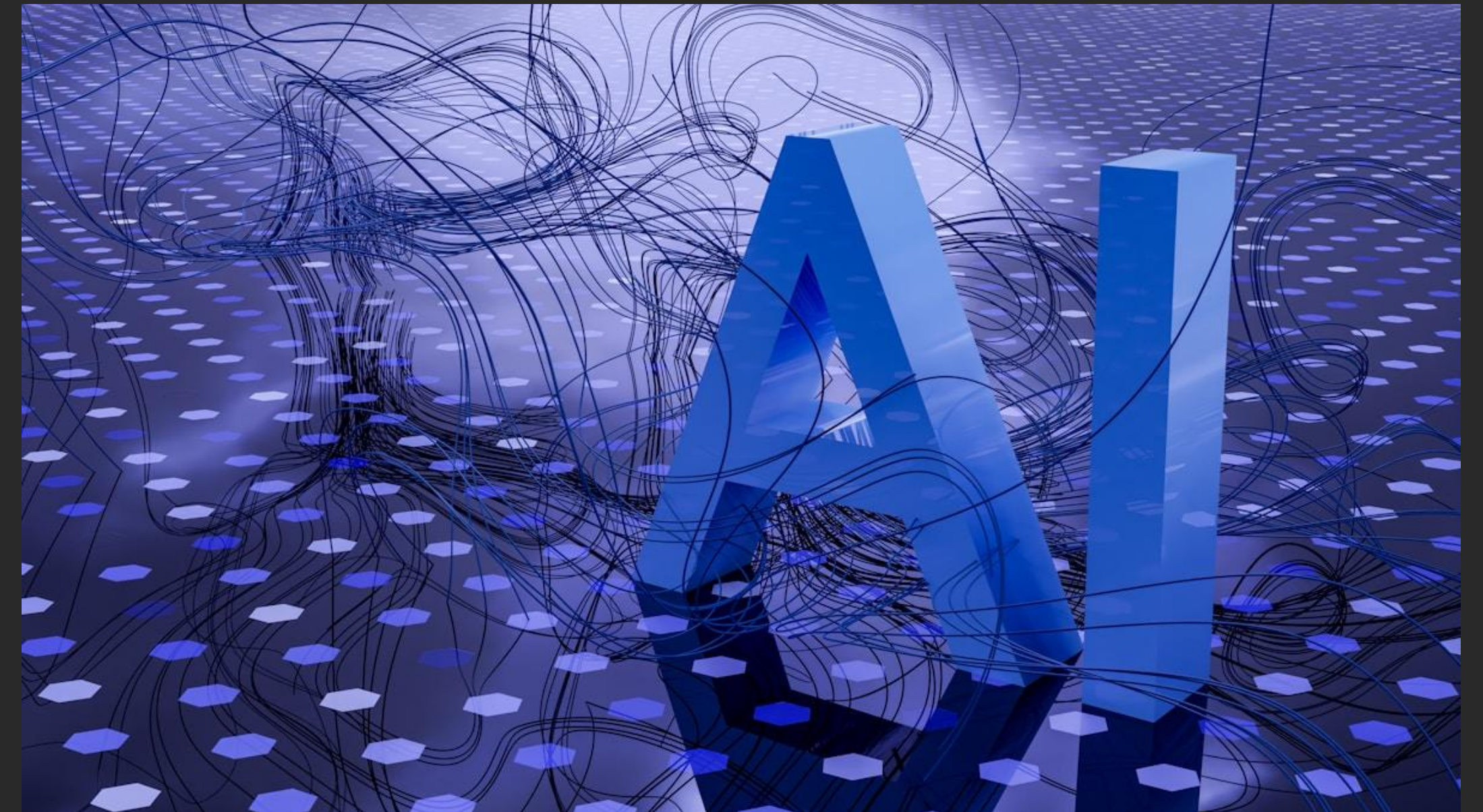
Individuals with hearing and speech impairments utilize sign language as their primary means of communication — both among themselves and with others within society.



- Sign language relies on a combination of manual movements, gestures, and facial expressions, forming a comprehensive linguistic system.
- Spoken and written languages dominate daily communication, creating a disparity that limits Deaf individuals' integration into society.
- This disparity creates a communicative gap between Deaf individuals and the rest of society, potentially limiting their ability to fully integrate.
- Advancements in AI and computer vision make it possible to develop intelligent systems capable of interpreting sign language and translating it into text or speech.

AI-Powered Solutions for Sign Language

Sign Language Recognition Systems represent one of the most significant applications in the field of human-computer interaction.



These systems function by analyzing hand movements and gestures using advanced techniques such as machine learning and deep learning. The field has witnessed remarkable progress with studies focusing on American Sign Language (ASL), Chinese Sign Language, and more.

Challenges in Sign Language Recognition

Five critical barriers preventing effective sign language translation systems



1

Time-based nature of data

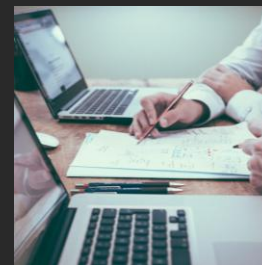
The signal relies on a chronological sequence of movements, requiring models capable of understanding time and not just the image.



2

Data imbalance

Some categories contain fewer samples than others.



3

visual noise

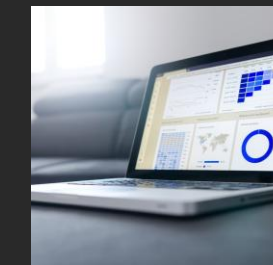
- Complex backgrounds
- Inconsistent lighting
- Image blurring



4

Differences among performers

Everyone performs the gesture in a different way, which increases the complexity of learning.



5

High accounting cost

- Video processing requires:
- High memory
 - High processing power

Project Goals & Deliverables

Four strategic objectives to bridge the communication gap through AI-powered sign language recognition.

- **KEYPOINT-BASED RECOGNITION**

Building a robust system using MediaPipe Holistic to extract skeletal landmarks from videos, ensuring privacy and processing efficiency.

- **TEMPORAL SEQUENCE MODELING**

Implementing Bi-LSTM architectures to capture the dynamic movement patterns and temporal dependencies of 100 different sign language classes.

- **DATA AUGMENTATION & BALANCING**

Developing a balanced dataset pipeline with targeted sample counts and advanced augmentation to improve model generalization across various signers.

- **ADVANCED AI RESEARCH**

Exploring cutting-edge improvements such as Attention Mechanisms and GCNs to enhance spatial-temporal feature extraction and accuracy..

References Studies in 2025

Dataset	Language	Size	Model	Accuracy
LSC-54	Colombian	26,000 images	Alphabet signs, controlled env	—
AI Hand Gesture	Multi	Large-scale, 3D landmarks	CNN	97%
Isharah	Saudi	30,000+ video clips	Continuous recognition	91%
TurkSign446	Turkish	446 words, static + dynamic	MediaPipe + RNN/LSTM/GRU	96%
SignAlphaSet	ASL	26,000 images + 300 videos	LSTM for sequence analysis	98%

WLASL100 Dataset

Word-Level American Sign Language Dataset



Dataset Source:

[Kaggle — WLASL100 Dataset](#)

Key Features:

- 100 ASL word classes
- Video-based format with diverse signers
- Real-world recording environments
- Suitable for continuous sign language recognition

Data Processing Pipeline

Input Encoding: Video frames normalized and resized. Temporal sequences preserved. LabelEncoder creates numeric class labels (labels_encoded.csv).

MediaPipe Holistic: Extracts 21 hand landmarks per hand + body pose + face mesh. Creates compact 258-dim feature vectors per frame.

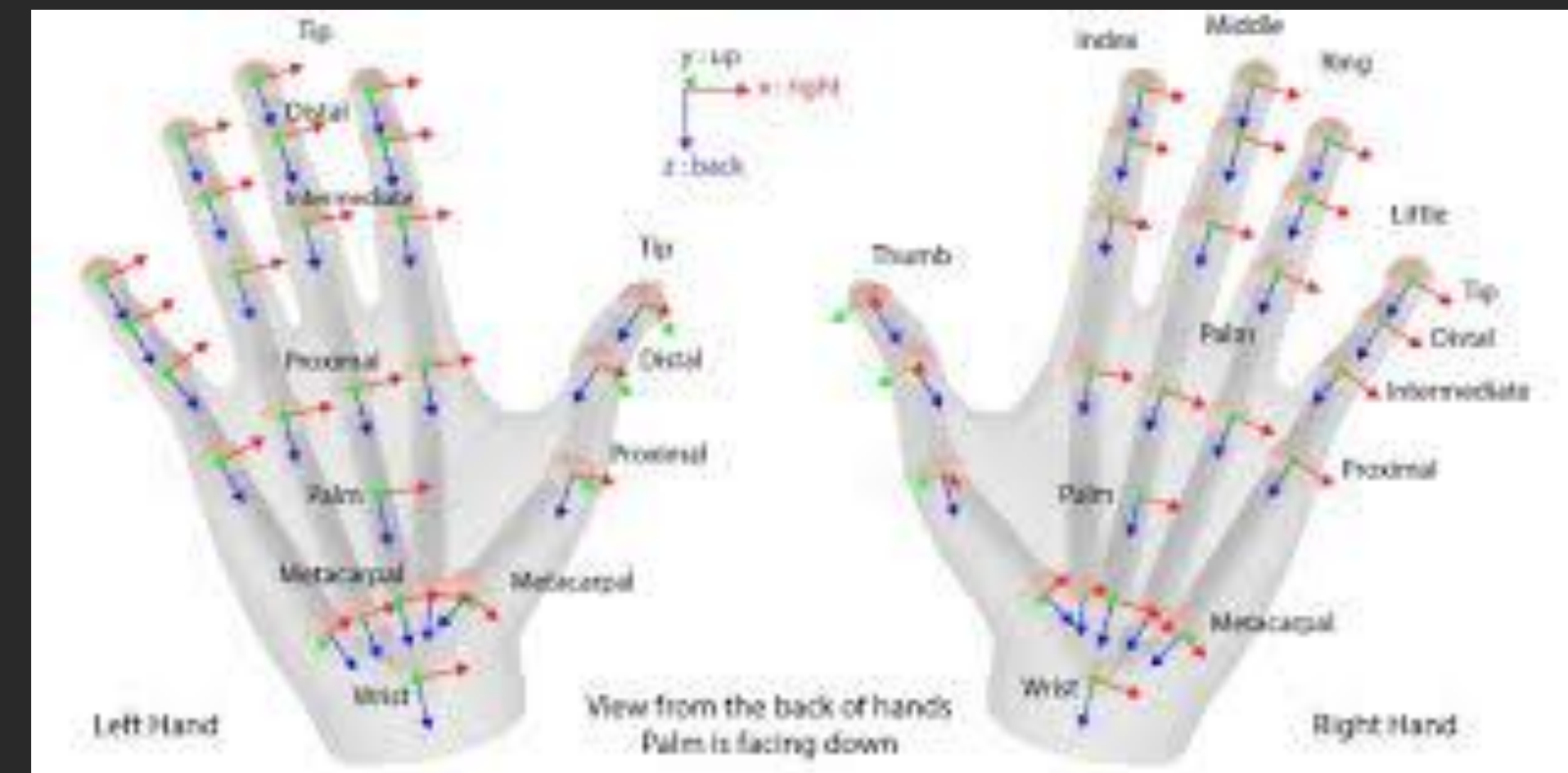
Augmentation: Rotation, scaling, flipping, time warping, random frame dropping. Increases diversity and model robustness.

Pose Landmarks
33 points x 4
values = **132**

Left Hand
21 points x 3
values = **63**

Right Hand
21 points x 3
values = **63**

Total: 132 + 63 + 63 = 258 features per frame

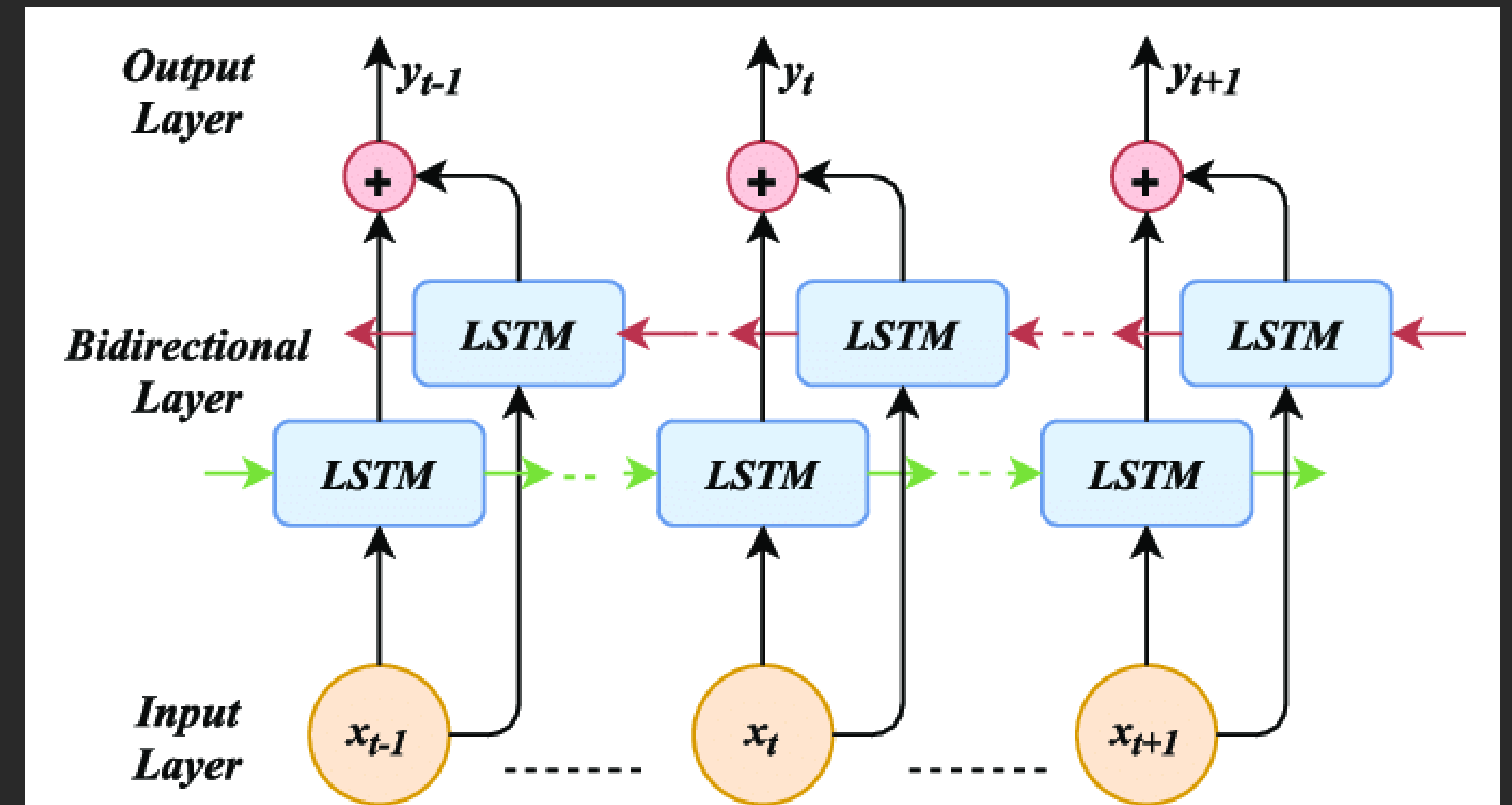


Work Plan

- **Feature Extraction** —MediaPipe Holistic extract visual patterns and body/hand landmarks from input frames.
- **Classification** — Bi-LSTM layers process sequential landmark data, followed by fully connected layers that classify gestures into categories.

Bi-LSTM Model Architecture

Bidirectional LSTM for sequential sign language recognition, processing temporal dependencies in both forward and backward directions.



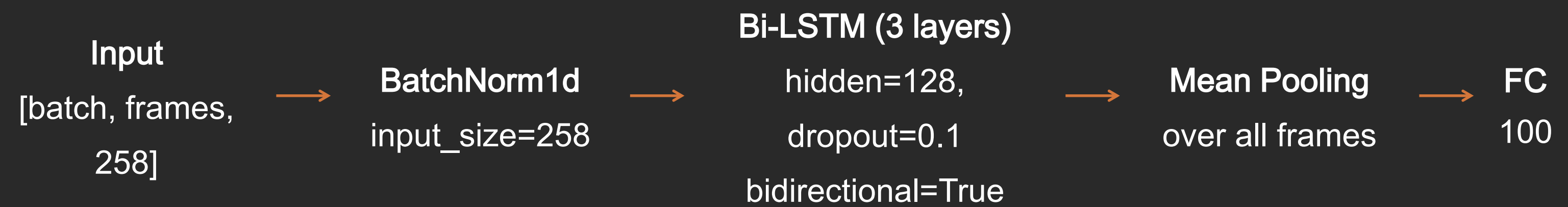
Architecture:

Input $\rightarrow$ Bi-LSTM $\rightarrow$ FC $\rightarrow$ Output

Key Advantages:

- Captures past and future context
- Superior temporal sequence modeling
- Combined with MediaPipe features
- End-to-end trainable with high accuracy

Bi-LSTM Model Architecture



Architecture Details

Input: 258 features

33 pose x 4 + 21 left hand x 3 + 21 right hand x 3

BatchNorm1d: Normalizes input features across the batch for stable training

Bi-LSTM (3 layers):

- `hidden_size = 128`
- `dropout = 0.1` (between layers)
- `bidirectional = True`
- Output: 256-dim (128 x 2)

Mean Pooling:

Averages LSTM outputs across all frames instead of using only the last frame — captures richer temporal information

Classifier Head

Sequential Classifier:

1. Linear(256 -> 256) + ReLU + Dropout(0.1)
2. Linear(256 -> 128) + ReLU + Dropout(0.1)
3. Linear(128 -> 100)
4. Linear(256 -> 128) + ReLU + Dropout(0.1)

Output: 100 sign classes

Key Design Choices:

- Progressive dimension reduction
- Heavy dropout for regularization
- Mean pooling over last-frame

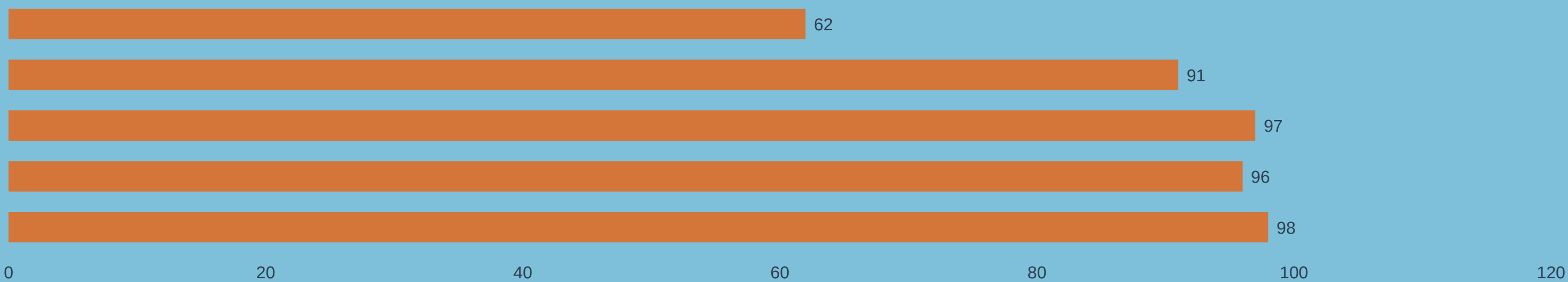
Training Results — 62% Validation Accuracy

62%

Validation Accuracy

On WLASL100 dataset
100 ASL word classes

Comparison with Related Work (2025)



Analysis & Key Factors

Why the gap with SOTA?

- **Real-world data complexity:** Varying backgrounds, unstable lighting, diverse signers
- **100-class challenge:** Fine-grained distinction between similar signs
- **Limited training data** per class in WLASL100
- SOTA results use controlled environments or smaller vocabularies

What 62% validates:

- End-to-end pipeline works correctly
- MediaPipe + Bi-LSTM is a **solid foundation**
- Mean pooling improves over last-frame approach
- Model generalizes to unseen real-world video data

Room for improvement through: larger datasets, attention mechanisms, transformer models

Conclusion & Future Work

Conclusions

Successfully built an **end-to-end sign language recognition pipeline** using **pipeline** using MediaPipe Holistic + Bi-LSTM achieving **62% accuracy** on **accuracy** on WLASL100.

Key contributions:

- Complete data processing pipeline with LabelEncoder
- 258-dim landmark feature extraction (pose + hands)
- Bidirectional LSTM with mean pooling
- Real-world video data processing

Future Work

- Increase **dataset size and diversity**
- Experiment with **attention mechanisms** and transformer models
- **GCN (Graph Networks)**: Better understand the natural connection between hand joints.
- Extend to **continuous sign language** recognition (not just isolated words)
- Add **Text-to-Sign** bidirectional translation



Thank You

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Any Questions?